

Web Development Course Outline (2 Months)

Duration: 2 Months

Classes: 5 per week

Class Duration: 1 Hour

Focus: JavaScript Advanced + Backend Basics (Node.js, Express, MongoDB) + Project

◆ WEEK 1 — Core JavaScript (Only What Is Needed)

Class 1: JavaScript Basics & Execution

- Variables (`var`, `let`, `const`)
- Data types
- How JavaScript runs
- Execution flow
- `strict` mode
- Clean coding habits
- Real-world usage of JavaScript

Class 2: DOM & Events

- Loops in js

Class 3: Functions & Logic

- Functions & arrow functions
- Parameters & return
- Scope (basic)
- Callbacks (intro)
- Logic-building practice

Class 4: Arrays & Objects

- Arrays & objects fundamentals
- Common array methods:
 - `push`, `pop`, `map`, `filter`, `forEach`
- Object structure & access
- Working with real data

Class 5: Asynchronous JavaScript (Intro)

- Synchronous vs asynchronous
 - `setTimeout`, `setInterval`
 - Promise concept (simple)
 - Why async code is important
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◆ PHASE 2 — React.js (Main Focus)

◆ WEEK 2 — React Fundamentals

Class 6: Introduction to React

- Why React?
- SPA concept
- React vs plain JavaScript
- Creating first React app
- Project structure overview

Class 7: Components & JSX

- Components concept
- JSX syntax
- Functional components
- Component reusability

Class 8: Props & State

- Props (data passing)
- State (`useState`)
- Updating UI dynamically
- Event handling in React

Class 9: Rendering & Lists

- Conditional rendering
- Rendering lists using `map`
- Keys in React
- Basic UI logic

Class 10: Hooks (Basic to Intermediate)

- `useState`
- `useEffect` (basic usage)
- Side effects in React

- Data fetching concept
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◆ WEEK 3 — React with Tailwind CSS & APIs

Class 11: Styling with Tailwind CSS

- Tailwind CSS overview
- Utility-first styling
- Responsive layouts
- Styling React components

Class 12: Forms & User Input

- Controlled inputs
- Form handling in React
- Validation (basic)
- Error messages & feedback

Class 13: Fetch API in React

- API concept
- Fetching data in React
- Displaying API data
- Loading & error handling

Class 14: Local Storage with React

- Using localStorage
- Saving app data
- Remembering user actions
- Practical examples

Class 15: Mini React Project

- React + Tailwind mini project
 - Component-based structure
 - State management
 - Real UI behavior
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◆ PHASE 3 — Backend Basics (Supportive)

◆ WEEK 4 — Node.js, Express & Integration

Class 16: Backend Basics & Node.js

- Frontend vs backend
- What is Node.js?
- Running backend code
- Simple server concept

Class 17: Express.js Essentials

- Express framework
- Creating routes
- Request & response
- Sending data to frontend

Class 18: REST API & MongoDB

- REST API concept
- HTTP methods
- MongoDB basics
- CRUD operations (intro)

Class 19: Authentication Concepts

- Login & signup flow
- Password hashing (concept)
- JWT overview
- Security basics

Class 20: Final Full Stack Project

Students will build a **complete basic-to-intermediate project**:

- Contact Management App
- Student Record System
- Login & Signup App
- CRUD Web Application

Tech Stack Used:

- JavaScript
- React.js
- Tailwind CSS
- Node.js
- Express
- MongoDB

